SARSA and Q-Learning Updates

We know Temporal Update goes like this:

$$\text{New Estimate} \leftarrow \text{Old Estimate} + \alpha[\text{Target} - \text{Old Estimate}]$$

For action-value functions:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[\text{TD Target} - Q(S_t, A_t)]$$

SARSA

- It stands for State Action Reward State Action.
- It updates the action-value function ($Q(S, A)$).
- It evaluates how good it is for an agent to take a specific action in a specific state, assuming the agent continues to follow its current policy.

Derivation:
SARSA is derived from the Bellman Expectation Equation:

$$q_\pi(s, a) = \mathbb{E}_\pi \left[R_{t+1} + \gamma q_\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a\right]$$

- SARSA samples the environment.
- Agent takes an action, observes R_{t+1}, lands in S_{t+1} and then uses its current policy to select A_{t+1}.

$$\text{TD Target}_{\text{SARSA}} = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})$$
$$\therefore Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\right]$$

Q-Learning

- Its goal is to directly approximate the optimal policy q^* instead of evaluating the current policy.

Q-learning is also derived from Bellman Expectation Equation:

$$q^*(s, a) = \mathbb{E} \left[R_{t+1} + \gamma \max_{a'} q_\pi(S_{t+1}, a') \mid S_t = s, A_t = a\right]$$

- Again we sample the environment.
- However, instead of waiting to see what action the agent actually takes next, we look at all possible actions in S_{t+1} and select one with the highest estimated Q-value.

$$\text{TD Target}_{\text{Q-Learning}} = R_{t+1} + \gamma \max_a Q(S_{t+1}, a)$$

$$\therefore Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)\right]$$

Comparison: SARSA vs. Q-Learning

Feature	**SARSA**	**Q-Learning**
Policy Type	Learns the value of the policy it is currently executing.	Learns the value of the optimal policy independently of the agent's actions.
TD Target	$R_{t+1} + \gamma Q(S_{t+1}, A_{t+1})$	$R_{t+1} + \gamma \max_a Q(S_{t+1}, a)$
Action Selection	Requires knowing the *actual* next action (A_{t+1}) before updating.	Only requires knowing the next state (S_{t+1}) to find the max action.
Risk	Learns safer policies if exploration is high.	Learns the optimal path even if it's dangerous during exploration.
Performance	Accumulates more reward during training because it avoids fatal failures.	Accumulates less reward during training if the optimal path is near severe penalties.
Convergence	Converges to the optimal policy *only if* the exploration rate (ϵ) slowly decays to 0.	Converges to the optimal policy as long as all state-action pairs are visited infinitely often.